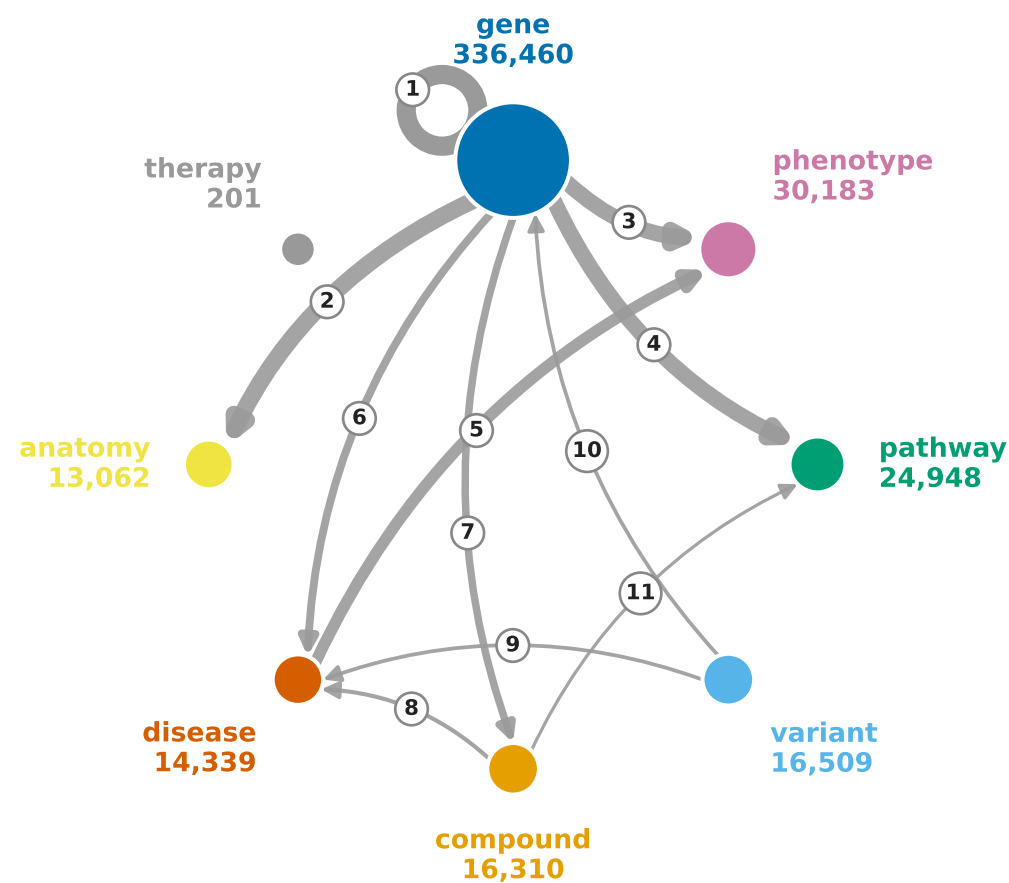


# Graph schema (metagraph): node categories and their relations — 452,012 nodes, 5,860,539 directed edges



## Relations (edge width $\propto$ log edge count)

- ① **gene → gene**  
Interacts With · 2,525,517 edges
- ② **gene → anatomy**  
Expressed In · 1,232,553 edges
- ③ **gene → phenotype**  
Has Phenotype · 858,859 edges
- ④ **gene → pathway**  
Actively Involved In · 838,866 edges
- ⑤ **disease → phenotype**  
Has Phenotype · 216,065 edges
- ⑥ **gene → disease**  
Gene Associated With Condition · 63,622 edges
- ⑦ **gene → compound**  
Interacts With · 56,333 edges
- ⑧ **compound → disease**  
Treats Or Applied Or Studied To Treat · 18,986 edges
- ⑨ **variant → disease**  
Causes · 16,855 edges
- ⑩ **variant → gene**  
Is Sequence Variant Of · 16,736 edges
- ⑪ **compound → pathway**  
Participates In · 14,081 edges